MANUSCRIPT REVIEW · LINE-LEVEL AUDIT

Line-Level Review of the Revised Categorical Viability-Weighted Curvature Framework

A granular audit of the 131-page revision (v3): eighty-one line-referenced findings spanning textual defects, mathematical flaws, internal inconsistencies, provenance signals, and validation-chain weaknesses, with a prioritized de-construction plan for the salvageable core.

Editorial Assessment Unit

Sequential read of all 7,897 text-layer lines · render-level verification of suspect passages · citation-coverage audit of 54 references · count-consistency audit across six framing layers

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1. Executive Summary

This report is a granular, line-level review of the revised 131-page manuscript “A Categorical Viability-Weighted Curvature Framework: From Autopoietic Sets to Non-Abelian Holonomy” (listed author “Z.ai”, draft dated 31 August 2026; 54 references; approximately 71,000 words). Every page of the text layer was read sequentially, with targeted render-level verification of suspect passages, a full citation-coverage audit of the bibliography against the body text, and visual inspection of the pages carrying typesetting anomalies. The review logged approximately eighty discrete, line-referenced findings, consolidated into the ledger in Section 10 and grouped into five categories: textual defects, mathematical flaws, internal inconsistencies, provenance signals, and methodological weaknesses in the validation chain.

The revision is substantially expanded relative to the 90-page version assessed on 30 August (v2): genome-scale biological validation on external expression and proteomics compendia has been added, prior-art references missing from v2 now appear in the bibliography, and several v2 errors (the optic-category source, the DV-versus-κ_V conflation, the inverse-limit terminology) have been corrected locally. These improvements are real, and the strongest new material, the transcript-versus-protein dissociation chain of Studies E24 to E27, is genuinely competent experimental science with honest confound controls. The revision, however, has been assembled as an append-only research log rather than rewritten as a paper, and the result is a manuscript that contradicts itself in public: six mutually different network counts, a limitations section that denies results the body headlines, proofs that state one thing while their lemmas assume another, and a body text that openly documents its own origin as an iterated response to AI audits and user requests.

The most consequential findings are these. First, the paper’s central quantity κ_V is three different objects at three different places (a geometric curvature contraction in Sections 2 to 5, an indicator-masked flux-rerouting statistic in Definition 3.21, and a time-course squared flux change in Studies E10 to E27), and the theory’s object never touches the data’s object. Second, the main definition of κ_V was changed after seeing the data, at an explicit in-text “user request”, “to test whether it strengthens the original findings”, and the promoted variant embeds the in-silico essentiality label inside the predictor. Third, eleven of the fifty-four references are never cited in the body, and at least six of those carry bibliography annotations falsely claiming “Cited in §4/§3/§21/§14/§5”, including the single most important prior-art attribution (Vereshchagin and Vitányi for the algorithmic rate-distortion distance). Fourth, the text retains session provenance throughout: “the user deposited”, “the user requested”, “the Qwen novelty assessment”, git commit hashes, build-log audits of dangling references, and a reference to “the source transcript”.

The editorial conclusion is that the manuscript is further from publishable than the v2 draft, not closer. The v2 problems (circular validation, uncited prior art, definitional instability) have not been removed; they have been multiplied and documented in the authors’ own words. The salvageable core is now clearer, however: the E24 to E27 biology chain, the fixed-model closure-test confusion matrices of E15 to E16, and the Phase III endpoint-sampling problem (currently buried under an incoherent ∞-categorical apparatus) are all genuinely interesting if extracted from the surrounding machine. Section 9 sets out a prioritized de-construction plan; Section 11 gives the verdict.

131	81	11	6
pages read line-by-line	line-referenced findings	references never cited in body	conflicting network counts

2. Scope, Method, and What Changed

The unit of review for this report is the line, not the claim. The full text layer (7,897 lines over 131 pages) was extracted with a layout-preserving converter and read sequentially from page 1 to page 131, with every theorem, definition, table, protocol, and numerical verdict logged against its line number. Where the extraction was ambiguous (superscripts flatten in text extraction, so 0.92^6 reads as "0.926"), the corresponding PDF page was rendered to an image and inspected visually before any finding was recorded; this report therefore distinguishes extraction artifacts (not flaws) from render-level defects (flaws), and several findings below were confirmed at the pixel level. Two audits were run over the whole document: a citation-coverage audit that parsed every bracketed reference group in the body and compared it against the fifty-four bibliography entries, and a count-consistency audit that extracted every stated count of biochemical networks, thresholds, and recovery horizons for cross-comparison.

The manuscript under review is the third identifiable version of the same work. Version v2 (90 pages, 41 references, 30 August) was the subject of the novelty assessment of 30 August. Version v3 (131 pages, 54 references, 31 August) incorporates, among other things: a new smooth-surrogate treatment of the algorithmic rate-distortion distance (Sections 4.1 and 4.2); genome-scale biological validation of an operational κ_V on the M3D, PRECISE, GSE64021, PaxDb and Schmidt proteomics datasets (Studies E24 to E27); a direct-raw-Keio essentiality test (E15) and an iML1515 cross-rebuild test (E16); a persistent-homology operationalization of the higher-categorical closure test (E4); nine additional designed metabolic networks (E through K-plus); and bibliography additions covering algorithmic rate-distortion, categorical cybernetics, categorical autopoiesis, chemical organization theory, network expansion, higher gauge theory and homotopy type theory. The scale of the expansion matters for the review: approximately 3,400 of the 7,897 lines are new content appended in "elevation rounds" (v1 through v20 of an internal study series), and the framing sections (abstract, introduction, roadmap, limitations) were not rewritten to match.

Aspect	v2 (30 Aug, 90 pp)	v3 (31 Aug, 131 pp)
Size and references	90 pages, 41 references, 15 numbered contributions	131 pages, 54 references, 16 numbered contributions; ~3,400 new lines in §19 alone
External data	None; all validation self-generated	M3D, PRECISE, GSE64021, PaxDb, Schmidt 2016 proteome, raw Keio supplements, iML1515, iAF1260, iMM904
Prior art	Vereshchagin–Vitányi, Hirota, Segura, Dittrich, Handorf, Fong–Spivak, Hedges all uncited	All added to the bibliography; but 11 of 54 never cited in the body, 6 with false "Cited in §X" annotations

Aspect	v2 (30 Aug, 90 pp)	v3 (31 Aug, 131 pp)
Framing integrity	Internally stale in places (roadmap counts)	Six mutually different network counts (2 / 4 / 5 / 10 / actual ~13); limitations contradict body results
Central definition	κ_V from Bregman divergence at dist_D	Three coexisting κ_V objects; FBA variant redefined post hoc (“v10 main definition”)
Provenance leakage	“Earlier drafts” references	“The user” (9 sites), “Qwen audit”, “source transcript”, commit 07e6d85, build-log audits, v1–v20 study rounds

Table 1. Structural comparison of the assessed versions. Line counts refer to the extracted text layer.

3. Textual and Typesetting Defects

Category A collects the defects that are visible on the rendered page without any mathematical interpretation: clipped text, a broken citation marker, cross-references to nonexistent sections, wrong enzyme-commission numbers, a biomass-units anomaly, and small numeric inconsistencies between restatements of the same quantity. These are individually small and collectively serious, because they show the PDF was never proofread end to end at the level of its own typography. A journal’s production team would return the file on the first three items alone; an editor should treat them as a proxy for the care taken with the less visible layers.

The gravest item is Definition 3.18, the five-step protocol that the abstract calls the paper’s operational centerpiece. In the rendered PDF the step labels are physically truncated: step (i) reads “Knockout”, but the following steps begin mid-word as “nder dynamics”, “above threshold”, “nstream cascade” and “toration control” (L780–791; confirmed by pixel-level inspection of page 14). The protocol is nevertheless cited by number throughout the paper, and readers must reconstruct from context that (ii) is a drop step, (iii) a recovery step, (iv) a downstream-cascade step and (v) a restoration-control step. The same overfull-box clipping reappears in Table 6, whose “likely missing term” column ships the fragments “growt” (three cells) and “regulat” (one cell; page 119). The one broken citation marker, “the standard essentiality threshold [?]” (L859), is an unresolved LaTeX \cite that survived into the final PDF; the in-text audit of v12 checked dangling \ref labels but evidently not \cite keys.

Cross-reference integrity fails repeatedly at the section level. The future-directions item 5 points to “the operational §1.4 form” (L7415), but Section 1 has no subsections. The authorship statement directs readers to “the manuscript’s §7 research-integrity notes” (L7715), but Section 7 is the composition-theorem section and contains no such notes. The §19 study rounds are cross-referenced in a numbering space (§§19.9 to 19.22) that collides with the Remark numbering of the same section (Remark 19.19 is the claim-by-claim verification, while “the v17 round (§19.19)” denotes the M3D study), so at least a dozen internal pointers are ambiguous. Two passages state the same statistic with different values: the iML1515 direct correlation to raw Keio essentiality appears as −0.018 in the E16 table (L5424) and −0.008 in the E18 re-run table (L5749), with the derived stability gap shifting between 0.103 and 0.093 accordingly.

Two factual slips have substantive weight. Glycogen phosphorylase is labeled EC 2.7.4.1 in the Network G design (L3363–3364), which is the EC number of polyphosphate kinase, correctly used for PPK in the same paragraph; glycogen phosphorylase is EC 2.4.1.1, so the same EC number is assigned to two different enzymes four lines apart. The attribution EC 2.6.1.12 to the invented aspartate–pyruvate transaminase M11 is also doubtful, since that EC entry is aspartate–prephenate aminotransferase. Separately, the wild-type biomass flux of iJO1366 is reported as 15.444 h^{-1} on 10 mmol glucose per gDW per hour (L5006–5013), roughly sixteen times the biologically possible growth rate on that uptake; the comparison with iML1515's 0.9259 is glossed as "different biomass stoichiometry" when the actual cause is almost certainly different units conventions (mmol-biomass versus gDW-normalized), which the relative-threshold logic papers over but never states.

4. Mathematical Flaws and Proof Gaps

Category B collects places where a stated theorem, lemma, or formula is wrong, is unproven as stated, or is proven at a level of rigor far below what its rhetorical use requires. The pattern is consistent: headline claims are stated as theorems "closing" the paper's own conjectures, while the proofs are sketches, restatements, or appeals to standard machinery that does not apply. Where the mathematical core is sound it is also elementary, and where it is not elementary it is not sound.

The smooth-surrogate chain of Section 4 is the clearest case. Definition 4.6 asserts that the surrogate $r_{\tau,\beta,D}$ is C^2, but its kernel is the squared positive part $[d-D]^2_+$, which is C^1 and not C^2 at the threshold locus $d = D$. Proposition 4.7 then compounds the error by claiming that the positive-part projection, the supremum over unit bivectors, and the maximum over margins "preserve smoothness", giving a "C^2 observable"; both the positive part and the supremum are nonsmooth operations, which is precisely why Section 4.2 needs Clarke subdifferentials three pages later. The paper contradicts itself on this point in adjacent subsections. Lemma 4.10's uniform Lipschitz bound is vacuous as proved: its own denominator lower bound $2^{-L/\tau}e^{-\beta \mathrm{diam}^2/\tau}$ tends to zero as $\tau \to 0$, and β is unbounded in the family, so no uniform bound is established; Theorem 4.11 calls the parameter set "discrete-enumerable" (L1095) when Definition 4.9 indexes it by a continuum, and its upper-semicomputability claim is asserted for a supremum over uncountably many functions. Remark 4.12's "consistency with dist_D up to O(1)" establishes only a lower bound on the envelope and no upper bound at all.

Corollary 4.14 is the single worst mathematical passage. It asserts, as a corollary, exactly the equality between the viability depth D_V and the curvature κ_V that Definition 2.1 had explicitly disclaimed as "a model-specific relation, not an identity"; its proof computes only the trivial radial average $\langle V_{max} - V \circ \gamma_a \rangle$ and never derives κ_V from the curvature 2-form; and its closing substitution "$F = \kappa_V$ and area = πa^2" gives $\kappa_V \cdot \pi a^2 = \pi a^4$, not the πa^2 the text claims (L1150–1177). The stratified-holonomy section has two further analytic defects: the matching condition (O3) is presented as an exact identity in a log/commutator form that is a Baker–Campbell–Hausdorff truncation of the standard gauge relation, and the small-loop formula (5) posits a loop "crossing the boundary once", which a closed transverse loop cannot do (the paper's own numerics use two crossings), while adding an order-one boundary reset term to an ε^2 expansion whose own verification exhibits an order-ε boundary contribution instead (L454–466).

Two “theorems” added in the elevation rounds are incorrect as stated. Theorem A of Study E13 (“maxRAF = terminal coalgebra”) defines $\Phi(S) = \{r \in U$: catalyzed by S, food-generated$\}$ and asserts $\Phi(S) \subseteq S$ for all S, which is false for that definition (a reaction catalyzed by S need not be in S); it further calls Φ “a polynomial endofunctor”, which the catalysis and closure conditions are not. The deflationary operator that would make the statement true is the standard greatest-fixed-point iteration known in the RAF literature, so nothing new is proved even after repair. Theorem B of the same study claims uniqueness of the functorial realization “by Strachey–Reynolds parametricity restricted to typed polynomial optics” (L5083–5084); parametricity is a property of polymorphic terms in typed calculi and cannot decide uniqueness of functors between the categories at hand. Corollary 17.3, the “univalence-identified fixed point”, imposes a metric on a simplicial mapping space, speaks of “a unique fixed point in each hom-set”, and asserts that “the type of contractible ∞-groupoids is equivalent to the universe U”, which is false (that type is contractible, not universe-sized); the proof chains these non sequiturs (L2734–2750). Remark 20.2’s claim that the upper-bound language is “theorem-justified” then restates the domination as “κ_V bounded above by the envelope E”, a unit error (a derivative ratio bounded by a function value) that recurs when a curvature 2-form integral is “bounded” by a scalar ratio integral (L6812–6836).

Finally, the 3/2 fatigue “derivation” (Theorem 12.2) is honest arithmetic wrapped in a misleading title. The exponent follows from an Itô-isometry computation plus the modeling assumption $\sigma^2 = \nu \cdot a$ (path-length-proportional diffusion); with amplitude-independent noise the same computation yields an analytic a^1 correction instead, so the “derived” exponent is manufactured by the ansatz. The first-passage Lévy story is decoration that the proof never uses, the constant C_{fat} is not derived (κ is “a kernel constant of order unity” absorbing a factor of 2.3 between theory and fit), and the reported 95% confidence interval [1.471, 1.493] excludes the theoretical 3/2, which the text acknowledges and then summarizes as “verified to within 1.4%” (L2030–2057).

5. Internal Inconsistencies

Category C collects statements that contradict other statements in the same document. The dominant cause is structural: the manuscript grew by appending elevation rounds without ever reconciling the framing layers, so the abstract, the introduction’s roadmap, the limitations section, and the conclusion each snapshot a different draft state. The clearest symptom is the answer to the question “how many biochemical networks does the paper test?”, which receives six mutually different answers in six places, tabulated below. A reviewer who reads only the abstract (four networks, climaxing at Network E’s 82.8%) and then only the body (eleven networks A through K plus variants, climaxing at 52/52 = 100% “FULL AUTOPOIESIS”) is reading two different papers.

The stale-framing problem reaches its sharpest form in the limitations section. Item 9 of Section 21.3 states that the closure test “does NOT produce a fully autopoietic verdict, and designing a fully autopoietic network with realistic biology remains open” (L7092–7098), which is the v2 position; the body’s Section 18.11 declares Network K a 52/52 = 100% Phase I verdict and Section 18.13 stress-tests it. Item 8 of the same limitations list says the piecewise holonomy formula is “verified numerically on a two-stratum abelian (U(1)) base" with the non-abelian extension "left for future work" (L7087–7091), while Section 3.1 presents SO(3) and CO(3) verifications plus five loop topologies as already done. The conclusion's "ten real biochemical networks" (L7482) matches neither the abstract's four, the roadmap's two, the limitations' four, nor the actual count of roughly thirteen (A, B, C, D, E, F, G, H, I, J, K, K-plus, and the iJO1366 overlay).

A second family of inconsistencies is conceptual rather than editorial. The D_V-versus-κ_V distinction, carefully drawn in Definition 2.1 ("it is not a curvature 2-form… the equality is a model-specific relation, not an identity"), is collapsed by Corollary 4.14 (which asserts the equality for all radially symmetric V), by Definition 6.1 (which cites "κ_V of Definition 2.1 (viability depth)"), by Remark 10.1 ("follows from Proposition 4.4 on the same model", without the computation), and again by the conclusion. The inverse-limit-versus-filtered-colimit confusion that Section 16.1 explicitly corrects ("this is a filtered colimit, not an inverse limit") resurfaces in contribution 7 of the introduction, in the title and text of Proposition 16.7 (whose "projection arrows $R_{max} \to R_i$" are described as "the inclusions", which point the other way), in the interpretation paragraph, in the script name, and in the conclusion. The endofunctor claim is stated in the introduction ("a single endofunctor T on the optic category") and withdrawn in Remark 7.8 ("requires explicit functorial semantics… not established here"), yet the withdrawn form continues to circulate in the §17 headline and the conclusion's "realization-to-contraction functor", which is itself never defined anywhere in the paper.

The remaining contradictions are localized but each would draw a referee comment. Definition 2.5 defines optic objects as triples (M, C, R) while Theorem 16.3 treats them as pairs with residuals on morphisms, two different formalisms coexisting without reconciliation. The base manifold is renamed from Θ to B in Definition 3.1 but Θ returns in Proposition 4.4, Remark 4.15 and Theorem 3.14 (with a differently shaped tuple). The structure group is SO(2)/SO(3) in Definition 2.2, requires added Weyl structure before CO(r) is licensed in Remark 2.3, and is flatly "the Fisher–Rao stabilizer CO(2)/CO(3)" in Remark 2.4. Equation (37) carries a three-term deterministic decomposition (πa^2, $a \cdot \kappa_V(a) = a^3$, $C_{fat} a^{3/2}$) while Claim C fits a two-term model; at the operating amplitude $a = 0.3$ the omitted a^3 term is 3.3 times the fitted fatigue term, and Section 11's stress test uses the three-term mean. The §15 "control" experiment "replaces" the RPSI optic f_2 with an expansion (determinant 1.15) when §8's canonical instantiation already defines $f_2 = 1.15 \cdot \mathrm{Id}$ as the expansion, making the control identical to the canonical setup. And the Network A results table marks component g "HOMEOSTATIC" with a recovery value of 164.2 (far above the 0.1 threshold) because it never dropped below threshold, while Remark 18.3 explains g's failure as a collapsed repair pathway, contradicting the table's own numbers.

Location (line)	Stated count	What is actually present
L238 (roadmap)	"two real biochemical networks" (Section 18)	Stale v2 text
L40–47 (abstract)	"four real biochemical networks", then lists five (A–E)	Count and list disagree; E is synthetic
L167–180 (contribution 12)	"four", listing A, B, C, D and E	Same five-as-four mismatch
L2899–2901 (§18 opening)	"two real biochemical networks"	Stale v2 text again

Location (line)	Stated count	What is actually present
L7092–7098 (limitations)	“four”; “no fully autopoietic verdict” exists	Contradicts §18.11’s 52/52 = 100%
L7482 (conclusion)	“ten real biochemical networks”	Sixth distinct count
Actual body content	—	A, B, C, D, E, F, G, H, I, J, K, K^+, iJO1366-overlay (~13)

Table 2. The six mutually different answers to “how many networks does the paper test?”

6. Provenance and Integrity Signals

Category D collects the passages that document, in the manuscript’s own words, how it was produced. Read together they establish that the document is a lightly-edited research-session transcript: an artifact in which an external “user” requests analyses, an AI system runs them, external AI audits (a “Qwen novelty assessment” and a 15-page “Novelty Assessment Report” held in an external_audits/ folder) supply the criticism that drives each revision round, and the rounds are versioned v1 through v20 with git commits. The scientific content is often honest about this, which is to the authors’ credit as analysts; as a manuscript it is disqualifying, because no journal can publish a paper whose methods narrative is addressed to a prompter rather than a reader.

The explicit markers are unambiguous and numerous. Section 19 opens by collecting “nine elevation studies conducted in response to the Qwen novelty assessment of the manuscript (external audit)” and cites “the v3 results from commit 07e6d85” (L4015–4024). Study E15 reports that “the user deposited the raw primary supplementary tables from Baba et al. itself into the project repository (folder raw tomoya baba supp/)” (L5187–5190); E16, E17, E12 and the v11–v13 rounds each record that “the user requested” or “the user then requested” the next analysis (L5302, L5505, L5680, L5958, L5801); the E27 replication is “(user-directed)” (L6686). The v14b round corrects “two false statements” in the manuscript’s own prior text and retracts an earlier interpretation (L6071–6125). Remark 20.6 compares the paper’s proposition to “the source transcript’s curvature-survival equivalence”, acknowledged “by the source” (L6940–6941). And the v12 round embeds a complete build log: a dangling-\ref audit with LaTeX line numbers and the names of the patch scripts “patch_manuscript_v12.py” and “patch_elevation_pdf_v12.py” (L6015–6048).

The citation-coverage audit compounds the integrity problem. Eleven of the fifty-four bibliography entries are never cited anywhere in the body: [5] (Brunerie et al., the orphaned v2 mis-citation), [32], [34], [35], [36] (higher-gauge references whose annotations describe usage that never appears), and, most importantly, [42] Vereshchagin–Vitányi, [45] Hirota et al., [46] Segura, [49] Kirchhoff et al., [50] Becker et al. and [51] Bravetti et al. Each of those six carries an annotation asserting a citation site: “Cited in §4 for prior-art precedence of the algorithmic rate-distortion distance”, “Cited in §3 for prior categorical-autopoiesis formalization work”, and so on. No such citations exist. Definition 4.1 therefore still presents the algorithmic rate-distortion distance without attribution at the point of definition, exactly as in v2; the bibliography merely performs attribution. Since the annotations were evidently written to satisfy an audit’s demand for attribution, this pattern deserves an explicit integrity note in any editorial file.

The authorship statement itself conflicts with the documented workflow. It asserts that “all mathematical content, theorem statements, proofs, numerical experiments, and conclusions were drafted by the author with the assistance of the Z.ai large language model for stylistic and editorial polish; the model did not originate novel mathematical claims” (L7713–7220). The body text attributes the design of Networks J and K, the robustness sweep, the definition promotion, and the data deposits to “the user”, attributes the paper’s revision agenda to two external AI audits, and refers to a “source transcript”; the data-and-code availability URL is a public repository named “deepseek-highly-general”, which contradicts the named drafting model. The statement also cites nonexistent §7 research-integrity notes. Whatever the true division of labor, the statement as written cannot be reconciled with the paper it accompanies, and the journal ’s AI-use and authorship policies will require an accurate account rather than this one.

7. Methodological Weaknesses in the Validation Chain

Category E collects the structural problems in how evidence is produced and selected, as distinct from whether any individual number is right. Five patterns dominate: definition shopping, label leakage, self-consistency presented as validation, threshold placement, and a serial search whose final statistic is reported without correction. Each pattern is documented in the manuscript’s own text, which again reflects real analytical transparency and still invalidates the headline claims as stated.

Definition shopping is explicit. The indicator-weighted variant of the FBA κ_V was promoted from a v9 refinement candidate to “the main definition” after a comparison of four weighting variants, at the “user”’s request, “to test whether it strengthens the original findings” (L5680–5683). The stated effects of the promotion are that E12’s calibration correlation jumped from +0.370 to +0.938 and that the iML1515 correlation flipped from −0.008 to +0.466 (L867–877). The definition text itself is titled “v10 main definition, indicator-weighted” and cites its own justification studies in-line. Worse, the winning indicator mask ${}^{c}[\Delta b > 0.05{\cdot}b_{wt}]$ is built from the in-silico essentiality criterion, so the promoted predictor contains the in-silico version of the outcome label; the paper concedes the semantics (“essentially asks given that a gene is in-silico-essential, what is its κ_V?”, L5646–5647), and the honest cross-rebuild evidence shows what the unmasked metric is worth on its own: below-chance ROC AUC 0.428 and top-10 precision 0.000 on iML1515, with the admission that “κ_V as currently defined (sum of squared flux changes) measures flux rerouting, not biomass reduction” (L5412–5445).

The self-validation pattern persists from v2 at every level. Claims A to E fit data simulated from the same formulas, with the fitted coefficient’s “prediction” being the data-generating parameter set in the same table (Claim C: predicted c_2 = 0.0500 against the simulator’s C_{fat} = 0.05; Claim D: K_{pred} = 25 versus K_{obs} = 25 with relative error 0.00). Claims F and G are deterministic simulations of textbook identities (so(3) commutators; the quantum-Zeno τ^2 expansion), and Remark 14.7 defends the resulting R^2 = 1.0000 by conceding that the simulated curve is the analytic law itself. The genome-scale “external” validations are internal in the way that matters: Study E12's held-out AUC of 0.953 correlates two statistics of the same knockout-FBA solution, and the genuinely external anchors are much weaker when tested directly (raw-Keio AUC 0.713, held-out specificity 0.180, MCC 0.085). The E14 "benchmark" against network expansion and chemical organization theory compares instruments run on different inputs (full model versus a 28-metabolite subnetwork) answering different questions, and then declares the dynamical test "strictly stronger".

The transcript-layer biology (E10 through E27) is the exception that proves the rule: its final numbers are defensible exactly because the round structure is honest. But the same round structure is a garden of forking paths when read as hypothesis testing. The abstract's $r = +0.374$ at $n = 433$ is the survivor of seven documented restructurings of the same question (E10 $r = 0.010$; two masking variants at $p = 0.33$ and 0.43; a retracted subset; a mapping inversion that found exactly one shared gene; a cross-condition test at $n = 7$, $p = 0.18$; then the M3D compendium), and no multiplicity correction is applied or discussed at the level of the headline. Threshold placement has the same flavor in the network designs: acceptance margins of 0.498 against a 0.5 pathwise threshold, 0.401 against a 0.4 Phase III threshold, and 0.853 against 0.5, with the thresholds themselves hand-set ($x_{thresh} = 0.1$, $\varphi = 0.4$ versus 0.5, $\tau = 0.30$, $n = 2$ perturbed trajectories) and the recovery horizon doubled from $T = 200$ to $T = 500$ precisely for the designed-to-pass Network E.

Two architectural gaps complete the picture. First, the paper's central quantity is not one object: the geometric κ_V of Propositions 4.4 and 4.7 (a positive-part curvature contraction of a Bregman divergence at a smooth surrogate), the FBA-operational κ_V of Definition 3.21 (an indicator-masked sum of squared flux changes), and the time-course $\kappa_V(r, t) = (v_r(t) - v_r(T_1))^2$ of Studies E10 and E22 are three different constructions with different domains, and the empirical program never touches the first. Second, the smooth-surrogate pathway that connects them in theory is the one the elevation studies dismantled: E5's calibration constant for the surrogate is shape-dependent, sign-flips on real FBA-derived viability, and yields a negative (unphysical) κ_V when the synthetic constant is applied, with the v4 verdict "NOT TRANSFERABLE" directly contradicting the v2 claim "transferable to subsequent real-data applications" two pages earlier (L4353–4470).

8. What Genuinely Improved or Works

Credit where due, stated plainly. The revision fixed several v2 defects: the optic-category definition now cites Riley's "Categories of optics" rather than the HoTT seminar notes (the [5] mis-citation survives only as an orphaned bibliography entry); the Hordijk–Steel, Dittrich, Handorf–Ebenhöh, Fong–Spivak–Tuyéras, Hedges, Hirota, Segura, Lurie, Riehl–Verity and Vereshchagin–Vitányi entries all exist in the bibliography now; the D_V versus κ_V distinction is at least drawn once (Definition 2.1) before being collapsed; the CO(r) structure group is correctly demoted to an extra modeling commitment in Remark 2.3; and the closure-test definition now includes a genuine viability threshold to defeat the trivial reinsertion failure mode. The methodology scaffolding around the prototype (calibration/holdout split, no-loop-drift control, equal-exposure permutation test, subdivision consistency, bootstrap variance) is sound experimental design in form.

The strongest new content is the E24 to E27 chain, which this review deliberately separates from the surrounding apparatus. The M3D genome-panel test ($r = +0.374$, $n = 433$, partial $r = +0.251$ controlling expression level, with the confound honestly described as “not wholly a level effect” rather than “unconfounded”), the 2×2 platform-by-class disambiguation on a common gene set with all four Fisher contrasts significant, the revision of the interpretation to growth-state-transition severity after non-carbon stress controls, and the Schmidt-quantitative-proteome replication establishing that the κ_V-graded organization is transcript-layer-specific while protein magnitudes are comparable, together constitute a real, honestly-caveated correlational finding. The E15 direct-raw-Keio analysis (transitivity gap, medium-mismatch mechanism, PEC confidence stratification) and the E16 iJO1366-versus-iML1515 model-gap comparison (30 to 13 gap candidates, 14 aminoacyl-tRNA synthetases resolved by the newer model’s explicit tRNA charging) are likewise publishable analyses in their own right. The v14b retraction, the honest nulls at every round, and the E5 arc that ends in its own constant’s refutation show authors capable of the right scientific instincts.

Finally, the closure-test program contains one genuinely good idea wrapped in two layers of apparatus. The knockout–collapse–threshold-crossing-recovery protocol with a downstream-cascade check and a restoration control is a legitimate operationalization of autopoietic closure, and its application to fixed, unengineered models (28/50 on iJO1366; 20/50 on both iAF1260 and iMM904, with honest cross-model agreement statistics) is the right experiment. The endpoint-sampling problem it discovered, that limit-cycle recoveries are scored by oscillation phase at the sampling instant, is a real methodological observation; the correct fix is period-averaging or multi-time sampling, both of which are one-line changes to the protocol. That the paper instead built an ∞-categorical “univalence-corrected” reinterpretation, operationalized as two perturbed reruns with a 30% tolerance, is a measure of how far the framing has drifted from the science.

9. Prioritized Opportunities for Improvement

The improvements below are ordered by expected effect on publishability. The first three are de-construction steps that reduce the manuscript to what its evidence can actually support; the remaining three are constructive steps that would give the salvageable core a defensible home. None of them requires new experiments beyond those already performed; all of them require deleting or relocating content, which is the operation the revision cycles have systematically avoided.

1. Split the single manuscript into three papers

The current document contains at least three separable works at three different credibility levels. The first is a methods paper on the dynamical closure test applied to fixed genome-scale models: Definition 3.18 (repaired typographically), the E2/E11/E15/E16 fixed-model results, the dep-ratio analysis, and the network-expansion and chemical-organization comparison run on matched inputs. The second is a short empirical report on the κ_V–transcript association: E22's panel construction, E24 to E27, with the metric renamed honestly (a model-derived flux-perturbation sensitivity, not a curvature) and the full search path either disclosed in a methods paragraph or pre-registered going forward. The third is the theoretical framework, which requires the mathematical repairs of the next item before it can be submitted anywhere. Each of the first two could be reviewed on its merits; none can while embedded in the current 131-page compilation.

The splitting also forces the naming problem to a resolution. Three constructs currently share the name κ_V; after the split, the geometric object keeps the name (or a better one), the FBA statistic becomes κ_{flux} or a rerouting index, and the time-course statistic is named for what it is. A reader must never be able to slide from a theorem about one to a correlation about another, which is what the current text invites at every level, up to and including the abstract.

2. Repair or delete the broken mathematics before any resubmission

Five items are repairable with bounded effort, and each currently blocks a referee's approval: (i) replace the $[d–D]^2_+$ kernel with a smooth soft-threshold (for example a Student-t or logistic kernel) so that the C^2 claim becomes true, or weaken every smoothness claim to C^1 plus Clarke differentiability consistently; (ii) restate Lemma 4.10 with a bounded parameter family (compact (τ, β) ranges, enumerable L) so that both the uniform Lipschitz bound and the upper-semicomputability claim become provable, and restrict Theorem 4.11 to a countable dense subfamily with a continuity argument for the envelope; (iii) delete Corollary 4.14 or restate it as a model-specific computation with the correct area factor, and reconcile it with Definition 2.1's disclaimer; (iv) restate Theorem 3.5's small-loop formula for two crossings with the boundary transition expanded to the order it actually contributes, and write the matching condition in the standard gauge form; (v) either delete Corollary 17.3, Theorem 17.2's uniqueness clause, Theorem A/B of E13 and the §17 apparatus, or commission actual proofs, which in the case of the terminal-coalgebra statement means correcting Φ to the deflationary operator and citing the finite-lattice fixed-point literature it instantiates.

One further repair has outsized value: define the "realization functor R" that the paper invokes at least six times, or remove every claim that depends on it. As written, the bridge between the categorical fixed-point story and the numerical contraction experiments is a name, not a construction, and every "unconditional" and "explicit" attached to it is unsupported. If the seven-optic composite cannot be realized faithfully, the honest paper says so in one sentence and the Banach analysis is reported for the toy maps it actually covers.

3. Excise the session provenance, then re-verify the bibliography against the text

A submission-ready manuscript cannot contain “the user”, “the Qwen assessment”, “the Novelty Assessment Report”, “the v14b audit”, “the source transcript”, commit hashes, version labels (v1–v20, “v10 main definition”), build logs, patch-script names, or folder paths such as “raw tomoya baba supp/”. Every one of these is a find-and-replaceable marker today, but removing them is not cosmetic: the surrounding narrative (elevation rounds, claim-by-claim verification of an audit, future-direction items marked closed inside their own statement) is built on them and must be rewritten as methods, not as responses. The authorship and AI-assistance statement must then be rewritten to be true, and reconciled with the repository name and the documented workflow; the current statement’s claim that the model “did not originate novel mathematical claims” is not sustainable against the paper’s own text.

The bibliography needs a mechanical pass that the v12 audit did not perform: compile-time citation checking (every “[?]” is a build failure), plus a report of uncited entries. The eleven uncited references should either be cited at the point of use (the Vereshchagin–Vitányi citation belongs at Definition 4.1, Hirota and Segura belong in Section 3’s related-work paragraph, which currently has none) or removed; the false “Cited in §X” annotations must be deleted in either case. This is the single highest-integrity-value edit available, because the annotations currently perform attribution that the text does not deliver.

4. Fix the endpoint-sampling problem in the closure test and drop the higher-categorical patch

The Phase I endpoint test scores limit-cycle recoveries by the oscillation phase at the sampling instant; the paper itself demonstrates this with AcCoA and ALA. The principled fix is to average the post-recovery trajectory over one period (or to sample at multiple offsets and require the median above threshold), which preserves the falsifiable binary verdict, removes the 0.498-versus-0.5 threshold knife-edge, and makes the Phase III apparatus unnecessary. The persistent-homology test of Study E4 (Betti numbers of the trajectory point cloud) can be retained as an optional descriptor, but the paper should stop calling a two-perturbation rerun the contractibility of an ∞-groupoid, and “univalence” should not appear in the operational verdict of a knockout experiment. The 2×2 platform-by-class design of E25 shows the authors can do clean disambiguation; the same discipline applied here would close the question in a page.

If a permissive multi-level verdict is retained for any reason, the acceptance logic must be a fixed hierarchy declared before the experiments, not a disjunction of three levels whose thresholds were tuned to the observed fractions. The current construction, in which a system is “autopoietic at Phase III” whenever it fails Phase I in a specific oscillatory way, is criterion relaxation documented as a theorem.

5. Report the designed-network lineage as engineering, not discovery

Networks E through K-plus are a design-convergence sequence: each variant adds isozymes, changes synthesis substrates, adds a food species, or boosts rate constants after seeing the previous verdict, and the paper's own monotone-improvement table (E 82.8% to K 100%) is a changelog. Presented as engineering with declared parameters (all k_{cat} values, the T = 200 versus 500 recovery horizons, the food set expansions), this material is a legitimate specification-based stress test of the closure protocol: it shows which design features (isozymes with independent synthesis substrates, basal constitutive expression, storage buffers) convert homeostatic verdicts into autopoietic ones. Presented as "Propositions" with "STRICTLY GREATER" verdicts, it invites exactly the reading the Qwen audit gave it, and the iJO1366-overlay null (Δ = 0) already shows the architecture adds nothing at genome scale. The honest framing is one section: the closure test as a design-verification instrument, the lineage as its calibration curve.

6. Pre-register the next external-anchoring step and demote the in-silico headlines

The paper's strongest available external anchors are the raw Keio panels, the expression compendia, and the quantitative proteome, and the honest numbers on those anchors (AUC 0.713 direct; r = +0.374 and partial +0.251; protein-layer null) are the ones the abstract should carry, with the in-silico AUC 0.953 reported as a calibration diagnostic of two FBA-derived statistics, not as validation. Before any further round, a pre-analysis plan for the essentiality question (primary endpoint, single metric definition fixed in advance, multiplicity correction for the already-explored variant space, and the unmasked κ_{flux} reported alongside any masked variant) would convert the documented search into a confirmatory analysis. The E16 result that the unmasked metric is below chance on the denser model should be treated as the standing null hypothesis to beat, not as noise to be masked.

10. Consolidated Line-Level Findings Ledger

The ledger consolidates the review's granular findings. Line numbers refer to the 7,897-line extracted text layer of the 131-page PDF; page numbers in parentheses refer to the PDF's own pagination. Severity is graded Critical (blocks publication on its own), Major (requires correction before publication), Moderate (requires an explicit fix or disclosure), and Minor (copy-editing). Findings that were verified against the rendered page rather than the text layer are marked "render-verified". Extraction artifacts such as flattened superscripts were excluded from the ledger after pixel-level verification showed correct rendering.

Lines	Finding (Category A: textual and typesetting defects)	Severity
L780–791 (p.14)	Definition 3.18 step labels (ii)–(v) are clipped mid-word in the rendered PDF ("nder dynamics", "above threshold", "nstream cascade", "toration control"); render-verified. The centerpiece protocol is typographically broken.	Critical

Lines	Finding (Category A: textual and typesetting defects)	Severity
L859 (p.15)	Unresolved LaTeX citation ships as "[?]": "the standard essentiality threshold [?]"; render-verified, sole occurrence.	Major
L7127–7185 (p.119)	Table 6 "likely missing term" cells clipped ("growt" ×3, "regulat"); render-verified.	Major
L839–877	Draft version metadata embedded in text: Definition 3.21 titled "v10 main definition, indicator-weighted"; "v1–v9 elevation studies", "(v10, E18; v11, E19)".	Critical
L7415–716	Cross-reference to nonexistent "§1.4" ("the operational §1.4 form").	Moderate
L7715	Authorship statement cites "the manuscript's §7 research-integrity notes"; no such content exists in §7.	Major
L4872–4875, 5928–5930	§19 study-round cross-references (§§19.14–19.22) collide with the Remark numbering space (Remark 19.19 etc.); a dozen internal pointers ambiguous.	Moderate
L5424 vs L5749	Same statistic, two values: iML1515 direct Pearson $r = -0.018$ (E16 table) versus -0.008 (E18 re-run table); derived gaps 0.103 versus 0.093.	Moderate
L3363–3364	Glycogen phosphorylase labeled EC 2.7.4.1 (polyphosphate kinase's number, used for PPK correctly four lines later); actual EC 2.4.1.1. M11's EC 2.6.1.12 attribution also dubious.	Moderate
L5006–5013, 5319	iJO1366 WT biomass reported as 15.444 h^{-1} on 10 mmol glucose/gDW/h (≈16× biologically possible); iML1515 0.9259; glossed as "different biomass stoichiometry" (units conventions, unstated).	Moderate
L5694–5696	"only 3 compute κ_V from FBA…: E10, E12, E15 and E16" (four items after "3").	Minor
L7188–7190	Table 6 caption uses singular "the author's hypothesis" against consistent "we" elsewhere.	Minor
L364, 322–327	Notation clashes: $\Gamma = (M, R, E_\Gamma)$ reuses M and R with different meanings than the RAF tuple (X, R, F, c).	Minor
L1204–1206	Lagrangian (22) $L = E[d] + \lambda I$ introduced, never used again (orphaned equation).	Minor

Table 3. Category A: textual and typesetting defects (14 findings).

Lines	Finding (Category B: mathematical flaws and proof gaps)	Severity
L962–989	Definition 4.6/Proposition 4.7 C^2 claims false: $[d-D]^2_+$ is C^1 not C^2 at $d = D$; positive part and supremum do not "preserve smoothness"; contradicts §4.2's own Clarke machinery.	Major
L1026–1051	Lemma 4.10's uniform Lipschitz proof vacuous: its lower bound $2^{-L/\tau}e^{-\beta \mathrm{diam}^2/\tau} \to 0$ as $\tau \to 0$; β unbounded in the family.	Major
L1015–1022, 1092–1096	Theorem 4.11: family indexed by a continuum ($\tau, \beta, D \in \mathbb{R}_+$) but proof calls it "discrete-enumerable"; upper-semicomputability of an uncountable supremum asserted, unproven.	Major
L1114–1133	Remark 4.12's "E consistent with dist_D up to $O(1)$" is a non sequitur (lower bound only; no envelope upper bound); Levin's theorem uncited.	Major
L1150–1177	Corollary 4.14: asserts the $D_V = \kappa_V$ equality that Definition 2.1 disclaims; proof never derives κ_V from F; "$F = \kappa_V$, area $= \pi a^2$" gives πa^4, not πa^2.	Critical
L406–409, 419–421	Gluing matching condition (O3) is a BCH truncation presented as an exact gauge identity (standard form $A_i = g^{-1}A_j g + g^{-1}dg$).	Major
L454–466	Theorem 3.5 formula (5): "crossing the boundary once" impossible for closed transverse loops (own numerics: ncross = 2); $O(1)$ reset term in an ε^2 expansion, inconsistent with the paper's own $O(\varepsilon)$ boundary verification.	Major

Lines	Finding (Category B: mathematical flaws and proof gaps)	Severity
L384–388 vs 417–423	Remark 3.2 defers the 2-categorical development to future work; §3.1 immediately claims the full gluing theorem “closing Conjecture 21.1”. Direct contradiction.	Major
L427–446, 477–611	Theorem 3.4 proof by asserted Giraud axioms for stratified spaces; Remarks 3.6–3.10 numerics verify self-derived formulas on toy constant-curvature connections; “closes Conjecture 21.1 in ALL THREE regimes” is an overclaim.	Moderate
L1709–1714 vs 1267, 1829–1861	Validation circularity: Claim C’s predicted $c_2 = 0.0500$ is the simulator’s input $C_{fat} = 0.05$; Claims A–E fit data generated from the same laws (Claim D “rel err = 0.00”).	Critical
L1709–1714 vs 1267, 1878–1879	Eq (37) three-term decomposition (πa^2, a^3, $C_{fat}a^{3/2}$) vs Claim C’s two-term fit; at $a = 0.3$ the omitted a^3 term is 3.3× the fatigue term; §11 uses the three-term mean. Never reconciled.	Major
L1982–2057	Theorem 12.2: 3/2 exponent manufactured by the ansatz $\sigma^2 = \nu \cdot a$ (constant σ^2 gives analytic a^1); first-passage Lévy framing unused in proof; C_{fat} not derived (κ “order unity” absorbs 2.3×); theoretical 3/2 outside the reported 95% CI.	Moderate
L2734–2750	Corollary 17.3 incoherent: metric on simplicial mapping spaces; “unique fixed point in each hom-set”; “type of contractible ∞-groupoids equivalent to U” false under univalence.	Critical
L5053–5068	E13 Theorem A incorrect as stated: $\Phi(S) \subseteq S$ false for the defined Φ; Φ not polynomial; correct content is the standard finite-lattice greatest-fixed-point iteration.	Major
L5078–5091	E13 Theorem B uniqueness “by Strachey–Reynolds parametricity” inapplicable to functor uniqueness; “numerical illustration” is $\mathbb{Z}/4$ arithmetic; does not close the functorial-semantics problem it names.	Major
L4234–4256	E3 transfer bound $\tau_{Zeno} \geq 1/(1+\log_2 N)$ manufactured (four unjustified steps); verified with 45–180× slack (vacuous); “monotone direction-of-effect” is the chosen functional form.	Major
L6812–6836	Remark 20.2 “theorem-justified” upper bound: restated domination “$\kappa_V \leq E$” is a unit error (ratio vs function value); “$\int\Sigma F \leq \int\Sigma\kappa_V \leq \int\Sigma\kappa_V^{alg}$” mixes 2-form integrals with scalar ratios.	Major
L912–948	Proposition 4.4’s “proof” is a restatement; gauge invariance argued via the square-root embedding’s SO(n)-invariance (a different group than the GC-gauge).	Moderate
L1446–1503	Theorem 8.2: the “seven optics” instantiated as six identical generic resnet blocks + one linear map (no semantic connection to the arcs); “Bregman-regularized” is a misnomer (Euclidean projection, no Bregman divergence); bound looser than the trivial Lip(T).	Major
L1390, 1437, 6957–6960	The “realization functor R from endo-optics to endomaps” is invoked repeatedly (called “explicit” in §21.1) and never defined.	Major
L2383–2389, 2417–2420	§15 Prop 15.4’s “control” (f_2 replaced by an expansion) is identical to §8’s canonical instantiation where $f_2 = 1.15 \cdot \mathrm{Id}$ is already the expansion; Fig. 9 caption repeats the confusion.	Major
L2333–2336	Prop 15.1 states strong-monotonicity and cocoercivity hypotheses never used in its proof.	Minor
L2759–2798, 3519–3526	Phase III: recovery map equated to the seven-optic composite with no connecting construction; contractibility operationalized as $n = 2$ perturbed reruns, 30% tolerance, no test statistic; disjunctive acceptance makes the verdict strictly more permissive.	Major
L3954–3967	Prop 18.26 “HoTT verdict” defined to coincide with Phase I (witness triangle present iff PASS); “contractible ⇔ Phase I = 100%” true by construction, labeled “falsifiable”.	Major
L3968–4002	Prop 18.28 SO(3) holonomies hand-assigned ($\alpha = 1.0$ “canonical”, axes by fiat); “identity holonomy” = $\exp(\alpha T)\exp(-\alpha T) = I$; Network K’s actual dynamics never enter; Theorem 3.5 invoked decoratively.	Moderate
L729–746	Prop 3.16: “$2\sqrt{p_a}$ places Δ on the positive orthant of the unit sphere”; $\lvert 2\sqrt{p}\rvert = 2$ (radius-2 sphere).	Minor
L1614–1673, 2291–2311	Claims F/G “empirical verdicts” are deterministic simulations of textbook identities; Remark 14.7 concedes the Zeno curve is the analytic law (48) evaluated at machine precision.	Moderate

Lines	Finding (Category B: mathematical flaws and proof gaps)	Severity
L1380–1405	Theorem 7.4 proof conflates optic-composition associativity with the monoidal structure; Construction 7.1/Table 1 residuals are semantic descriptions (“residual = Fisher–Rao metric”), not objects of a concrete C; “compatibility $A_i = M_{i+1}$ by construction” asserted by declaration; O2 forward is a CPTP channel, not a Set-map.	Moderate

Table 4. Category B: mathematical flaws and proof gaps (28 findings).

Lines	Finding (Category C: internal inconsistencies)	Severity
L40–47, 167–180, 238, 2899, 7092–7098, 7482	Network count stated as 2 / 4 (with five listed) / 4 / 2 / 4 (“no fully autopoietic verdict”) / 10 across six framing sites, against ~13 actual networks (A–K^+ plus overlay); limitations deny the body’s 52/52 = 100% headline.	Critical
L246–274 vs 1150–1162, 1259, 1722–1724, 7454–7456	D_V vs κ_V separation drawn in Definition 2.1, then collapsed by Corollary 4.14, Definition 6.1 (“κ_V of Definition 2.1 (viability depth)”), Remark 10.1 and the conclusion.	Major
L135–137 vs 2457–2458, 2628–2631, 2644, 7407–7417, 7475	Inverse-limit vs filtered-colimit: contribution 7 says “inverse-limit construction”; §16.1 explicitly corrects it; Prop 16.7 item 2 titled “Inverse limit” with “projection arrows” described as inclusions (pointing the wrong way); conclusion repeats “inverse limit”.	Major
L80–81 vs 1426–1435	Intro claims “a single endofunctor T on Optic(C)”; Remark 7.8 concedes the endofunctor semantics is “not established here” (typed endo-optic only); withdrawn form persists in §17 and conclusion.	Major
L275–312 vs 284–286	Structure group triple-claim: prototypes use SO(2)/SO(3) (Def 2.2); CO(r) needs added Weyl structure (Remark 2.3); “the Fisher–Rao stabilizer is CO(2)/CO(3)” flatly (Remark 2.4); spherical FR point stabilizer is O(n−1).	Moderate
L316–319 vs 2489–2491	Optic objects as triples (M, C, R) in Definition 2.5 versus pairs with morphism-residuals in Theorem 16.3; two formalisms coexist without reconciliation.	Moderate
L353–361 vs 676, 931, 1178	Base manifold renamed Θ → B in Definition 3.1; Θ returns in Prop 4.4, Remark 4.15, Theorem 3.14 (with a differently shaped tuple).	Minor
L3034–3039 vs 2859–2873, 3055–3060	§18.4’s written FBA protocol (knock out ALL producers of m) cannot generate its own nprod-stratified results; implementation must have knocked out only active producers. Text vs code mismatch.	Major
L2936–2968 vs 3408–3412, 3519–3526	Network A table: g marked HOMEOSTATIC with recovery 164.2 (never dropped below threshold); Remark 18.3 explains g as a collapsed repair pathway, contradicting the table.	Moderate
L3413	PYR “baseline 0.0, recover 5.7, causally internal”: a component whose baseline is below the viability threshold passes the closure test; drop-condition semantics break.	Moderate
L3186 vs 2912	Recovery horizon T = 200 for Networks A/B/D but T = 500 for the designed-to-pass Network E; goalpost move; §18.1 also calls mass-action kinetics “Michaelis–Menten–type” (no Km, no saturation) while E uses true MM.	Moderate
L3352–3354 vs 4479 vs robustness tables	“Substrate MAL is food (stable at 100)” in the E6 design contradicts MAL being a tested non-food intermediate (fragile 0.99 in K^+).	Moderate
L1830–1831 vs 2139–2145	Claim E acceptance criterion drifts: n = 3 “$T_{ctrl}/T_{loop} \geq 5$” (Def 6.1, Table 2) vs n = 4 “$T_{loop} < 2$, $T_{ctrl} > 1$” (Table 3).	Moderate

Lines	Finding (Category C: internal inconsistencies)	Severity
L7245–7267 vs 7087–7091 vs §3.1	Non-abelian verification status stated three ways: Conjecture 21.1 restatement (U(1) only), Limitation 8 (non-abelian "future work"), §3.1 (SO(3), CO(3), five topologies done).	Moderate
L3621–3626, 3653–3658, 3724–3732	Acceptance margins cluster at thresholds: 0.498 vs 0.5; 0.401 vs 0.4; 0.853 vs 0.5; thresholds hand-set (φ = 0.4 vs 0.5, τ = 0.30) and appear placed around observed values.	Major
L2103–2145	Table 3 stray "+0.99958" in the c_{comm} row (commutator-fit R^2 mangled into the wrong column).	Minor
L3672–3675 vs §18 structure	Network J has no subsection; introduced parenthetically inside §18.11's opening; section numbering skips I → K with J in passing.	Moderate

Table 5. Category C: internal inconsistencies (17 findings).

Lines	Finding (Category D: provenance and integrity signals)	Severity
L4015–4024	§19 "Novelty-Assessment Elevation Studies": "nine elevation studies conducted in response to the Qwen novelty assessment (external audit)"; cites "commit 07e6d85"; v3/v4/v5 study versions; Table 4 organized by the audit's own section numbers.	Critical
L4978–4988	§19.8: responses to "the Novelty Assessment Report (editorial-style external audit, 15 pages, see external_audits/ folder)"; its three upgrade recommendations "closed" in-study.	Critical
L6071–6125	v14b round corrects the manuscript's own prior text: "two false statements (a nonexistent EcoCyc proxy… and the 'per-gene heterogeneous κ_V'/'real biological signal' interpretation)"; a previously reported subset finding retracted in-place.	Critical
L6940–6941	Remark 20.6 refers to "the source transcript's curvature-survival equivalence", "acknowledged by the source": a source transcript is cited as the origin of prior claims.	Critical
L3672–3673, 3853, 3896, 5187–5190, 5302, 5505, 5680, 5801, 5958, 6686	"The user" appears as the driver of design and analysis decisions throughout ("the user-suggested…", "the user deposited… (folder raw tomoya baba supp/)", "the user then requested…", "(user-directed)").	Critical
Bibliography [5], [32], [34]–[36], [42], [45], [46], [49]–[51]	Eleven of 54 references never cited in the body; six ([42] Vereshchagin–Vitányi, [45] Hirota, [46] Segura, [49] Kirchhoff, [50] Becker, [51] Bravetti) carry annotations falsely claiming "Cited in §4/§3/§21/§14/§5"; Definition 4.1 still presents $dist_D$ without attribution at the point of definition.	Critical
L6015–6048	Build log embedded: dangling-\ref audit (12 label namespaces, 321 labels, 704 references, LaTeX line numbers) and patch scripts by name (patch_manuscript_v12.py, patch_elevation_pdf_v12.py).	Critical
L4297	"The audit's §8.4 prescription to 'remove the HoTT section' is REJECTED in favor of elevation": referee-response negotiation in the body; L750/L1193 also reference "the audit".	Moderate
L2884–2895, 3430–3434, 3553–3555, 3846–3847, 5293–5298, 7393–7406	The manuscript implements its own §21 future-directions list within the same document ("the future-direction item of Section 21 had suggested X; the present subsection implements that suggestion"); future work marked [CLOSED] inside its own statement.	Critical
L7713–7720	AI-assistance statement ("the model did not originate novel mathematical claims") contradicted by the documented workflow; code availability URL is a public repository named "deepseek-highly-general", contradicting the stated drafting model.	Critical

Lines	Finding (Category D: provenance and integrity signals)	Severity
L5187–5188, 6392–6393, 6152–6167	Session-environment details in the body: file deposits into the project repository, “the COLOMBOS host itself was unreachable from the analysis environment”, artifact names with “internet-access verification table”.	Moderate

Table 6. Category D: provenance and integrity signals (11 findings).

Lines	Finding (Category E: methodological weaknesses)	Severity
L867–877, 5680–5683	Definition shopping documented in-text: the indicator-weighted κ_V promoted to “main definition” at the user’s request “to test whether it strengthens the original findings”; calibration r jumped +0.370 → +0.938; iML1515 flipped −0.008 → +0.466.	Critical
L5528–5529, 5646–5651	Label leakage: the indicator mask ᶜ[$\Delta b > 0.05 \cdot b_{wt}$] embeds the in-silico essentiality criterion in the predictor; the “Keio correlation” largely measures the model’s own essentiality accuracy (text concedes the conditional-on-essential semantics).	Critical
L5412–5457	Unmasked metric null on the denser model: iML1515 ROC AUC 0.428 (below chance), P@10 = 0.000, “the gap is now in the WRONG direction”; admission that κ_V “measures flux rerouting, not biomass reduction”.	Major
L4839–4855, 4871–4922	Garden of forking paths, transparently documented: the headline r = +0.374 (n = 433, $p \approx 10^{-15}$) is the survivor of seven restructurings (r = 0.010 → masks at p = 0.33/0.43 → retraction → mapping inversion (one shared gene) → n = 7, p = 0.18 → M3D); no multiplicity correction; abstract reports only the survivor.	Major
L4301–4470	E5 self-refutation arc: the surrogate’s calibration constant is shape-dependent (1.625/1.367/1.263), sign-flips on real FBA viability, and yields negative (unphysical) κ_V; v2’s “transferable to real-data applications” directly contradicted by v4’s “NOT TRANSFERABLE” two pages later.	Major
Props 4.4/4.7 vs Def 3.21 vs E10/E22	Three distinct objects named κ_V (geometric curvature contraction; indicator-masked FBA flux rerouting; time-course squared flux change). The empirical program (E10–E27) never touches the theoretical object; the abstract’s main proposition concerns the latter.	Critical
L5026–5047, 5029–5036	E12 “external validation” is FBA-versus-FBA: κ_V and the essentiality label are both statistics of the same KO-FBA solution (held-out AUC 0.953 in-silico); the direct external test (E15) gives AUC 0.713, specificity 0.180, MCC 0.085; “transitive prediction” framing.	Major
L5140–5166	E14 benchmark invalid as designed: dynamical test on full iJO1366 vs NE scope and a 28-metabolite COT subnetwork; different inputs and different questions (“can m be made?” vs “is m’s production causally necessary?”); “strictly stronger” is a category error; “no false positives” is circular.	Moderate
L3179–3186, 3284	Unexamined free parameters: $x_{thresh} = 0.1$, $C_{fat} = 0.05$ (hand-set), $\varphi = 0.4/0.5$, $\tau = 0.30$, $n = 2$, $k_{cat} \in \{0.5\ldots30\}$ (per-reaction overrides), Km = 0.1, $\delta = 0.05$; sensitivity analyzed for a subset only.	Moderate
L4417–4444 vs 4595–4632	E17/E18 cross-rebuild “stability” of the masked variant comes from the mask; E7’s universality table shows G/H/I rows identical (5/11, 0.450, 0.714) across three different networks and K (46%) below F/J (60%).	Moderate
L4583–4605, L4622–4646	Positive: honest negatives retained in tables (E5 “NOT TRANSFERABLE”; E9 “NO_TRANSITION”; E16 counter-finding; v14b retraction); the round structure discloses the search that produced the headline.	Minor

Table 7. Category E: methodological weaknesses (11 findings).

11. Editorial Verdict

The verdict of this line-level review is that the revised manuscript is not publishable and is further from publication than the version it revises. That conclusion is not primarily about the new biology, parts of which are sound, nor even about the mathematics, much of which is repairable. It is about the document as a document: an append-only compilation that preserves every superseded framing layer, contradicts itself on its own basic counts, embeds the transcript of its production, claims attributions in the bibliography that the text never makes, and states as theorems a set of propositions whose correct versions are either elementary or false. Each defect is individually documentable at a line reference; the aggregate is decisive.

For an editor, the practical reading is this. The theoretical core (Sections 2 to 8, 12, 16, 17) cannot be sent to referees in this state: five of its load-bearing statements are wrong or unproven as written, and the central object's name is shared by three incompatible constructions. The closure-test program (Section 18) is salvageable only as an engineering-verification study on fixed models, with the designed-network lineage re-framed and the higher-categorical verdict machinery deleted. The empirical biology (E22, E24 to E27, with E15 and E16) is the strongest material in the paper and could anchor a short, honest methods-and-findings report once the metric is renamed, the search path is disclosed, and the in-silico headlines are demoted to calibration diagnostics.

The three integrity items that would surface in any competent refereeing round, and that the authors should address before anything else, are the falsely annotated bibliography entries, the AI-assistance statement that the body text contradicts, and the in-text documentation of definition changes made after seeing the data. None of these requires new science to fix; all three require deletions and true statements. The recommendation is therefore not "revise and resubmit" in the ordinary sense but "deconstruct and re-submit as three separate, honestly framed works", following the plan in Section 9. The raw material for at least one publishable paper is present; it is currently buried under the machinery of its own production.